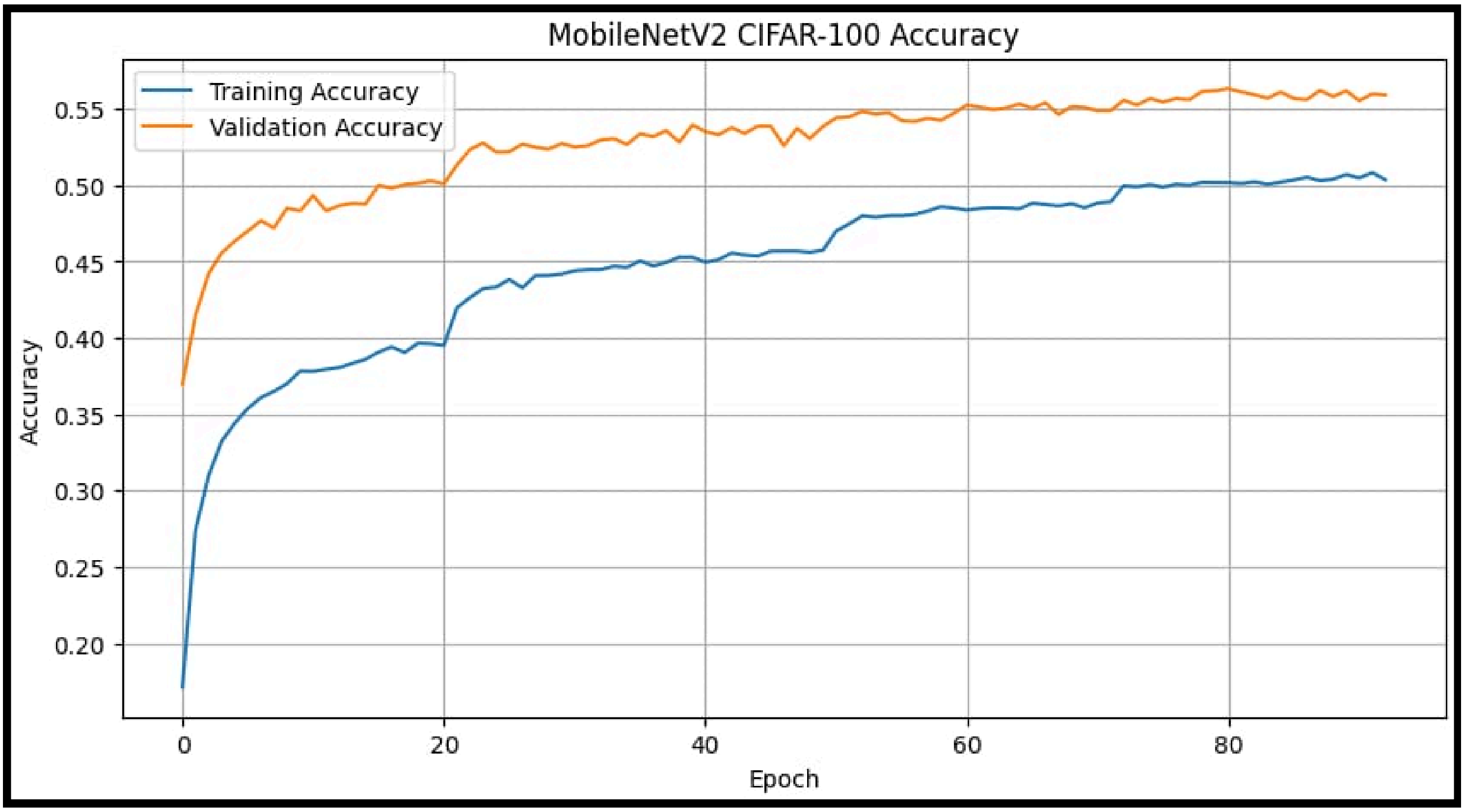
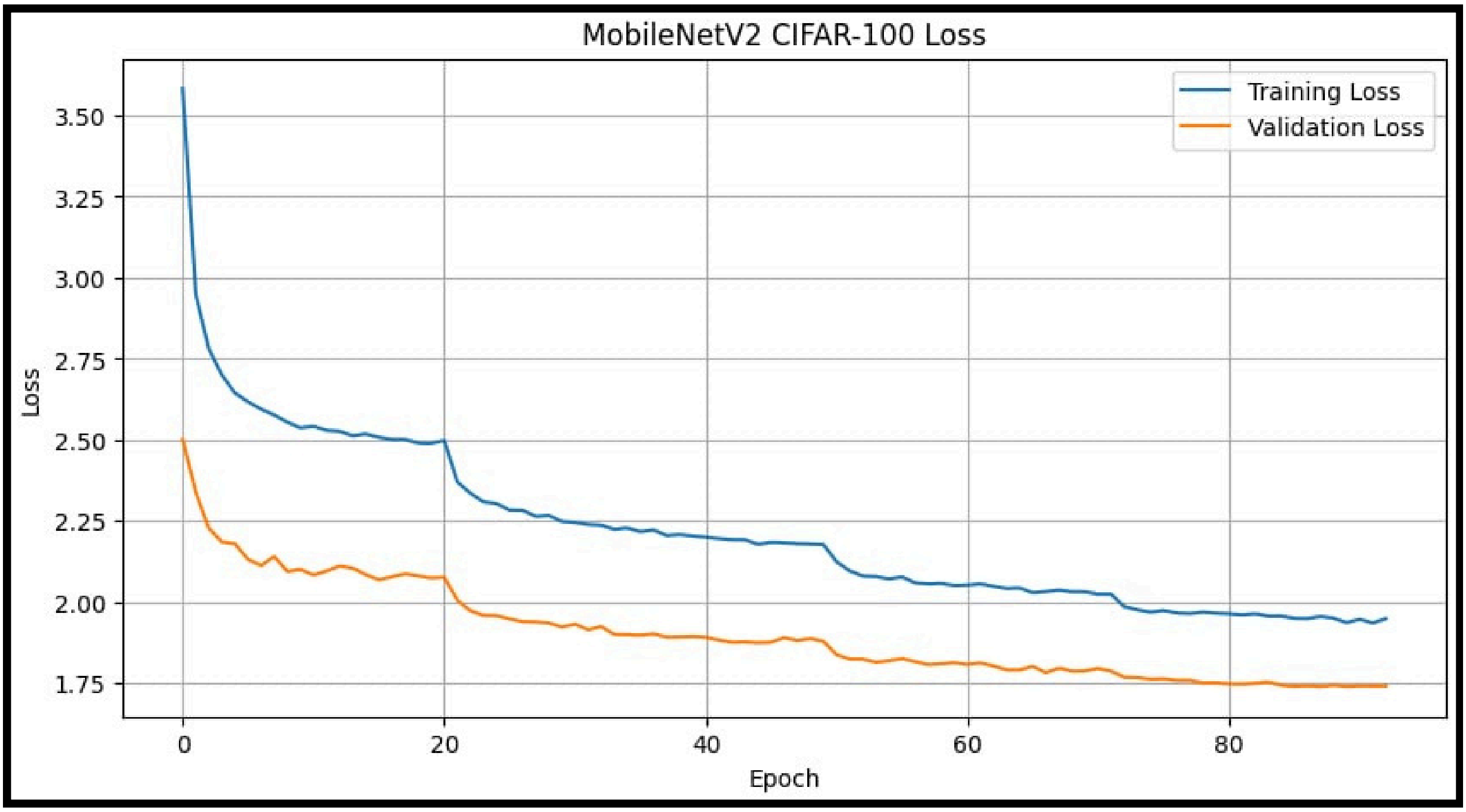
MOBILENETV2 CIFAR-100 DATASET

Visualizing the Learning Curves





MOBILENETV2 CIFAR-100 DATASET

#Classification Report

	precision	recall	f1-score	support
0	0.8056	0.8700	0.8365	100
1	0.6239	0.7300	0.6728	100
2	0.5053	0.4800	0.4923	100
3	0.3824	0.3900	0.3861	100
4	0.2349	0.3500	0.2811	100
5	0.4865	0.5400	0.5118	100
6	0.4857	0.5100	0.4976	100
7	0.7123	0.5200	0.6012	100
8	0.6972	0.7600	0.7273	100
9	0.7130	0.8200	0.7628	100
10	0.5529	0.4700	0.5081	100
11	0.3516	0.3200	0.3351	100
12	0.6863	0.7000	0.6931	100
13	0.5541	0.4100	0.4713	100
14	0.6395	0.5500	0.5914	100
15	0.3529	0.4200	0.3836	100
16	0.6857	0.7200	0.7024	100
17	0.5852	0.7900	0.6723	100
18	0.5532	0.5200	0.5361	100
19	0.6582	0.5200	0.5810	100
20	0.7333	0.7700	0.7512	100
21	0.5878	0.7700	0.6667	100
22	0.7300	0.7300	0.7300	100
23	0.5725	0.7500	0.6494	100
24	0.7156	0.7800	0.7464	100
25	0.5275	0.4800	0.5026	100
26	0.5618	0.5000	0.5291	100
27	0.3333	0.3800	0.3551	100
95	0.4480	0.5600	0.4978	100
96	0.3388	0.4100	0.3710	100
97	0.4444	0.6000	0.5106	100
98	0.3478	0.1600	0.2192	100
99	0.6952	0.7300	0.7122	100
accuracy			0.5648	10000
macro avg	0.5650	0.5648	0.5585	10000
weighted avg	0.5650	0.5648	0.5585	10000

PROJECT PERFORMANCE SUMMARY

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================================================================================
                          PROJECT PERFORMANCE SUMMARY
================================================================================

DATASET INFORMATION
--------------------------------------------------------------------------------
Dataset Name                  : CIFAR-100
Number of Classes             : 100
Training Samples              : 45,000
Validation Samples            : 5,000
Test Samples                  : 10,000

MODEL PERFORMANCE
--------------------------------------------------------------------------------
Training Accuracy             : 50.83%
Validation Accuracy           : 56.34%
Test Accuracy                 : 56.48%
Test Loss                     : 1.7365

PERFORMANCE METRICS
--------------------------------------------------------------------------------
Precision                     : 56.50%
Recall                        : 56.48%
F1 Score                      : 55.85%

MODEL INFORMATION
--------------------------------------------------------------------------------
Model Size                    : 18.50 MB
Model Name                    : MobileNetV2
Total Parameters              : 3,070,884
Trainable Parameters          : 812,900
Non-Trainable Parameters      : 2,257,984
```

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INFERENCE PERFORMANCE
--------------------------------------------------------------------------------
Total Inference Time          : 0.221922 sec
Average Time Per Image        : 0.002219 sec
Average Time Per Image (ms)   : 2.22 ms

TRAINING INFORMATION
--------------------------------------------------------------------------------
Optimizer                     : Adam
Fine-Tuning                   : SGD + Momentum
Batch Size                    : 64
Epochs Completed              : 93

PROJECT STATUS
--------------------------------------------------------------------------------
Transfer Learning             : Completed
Model Evaluation              : Completed
Fine-Tuning                   : Completed
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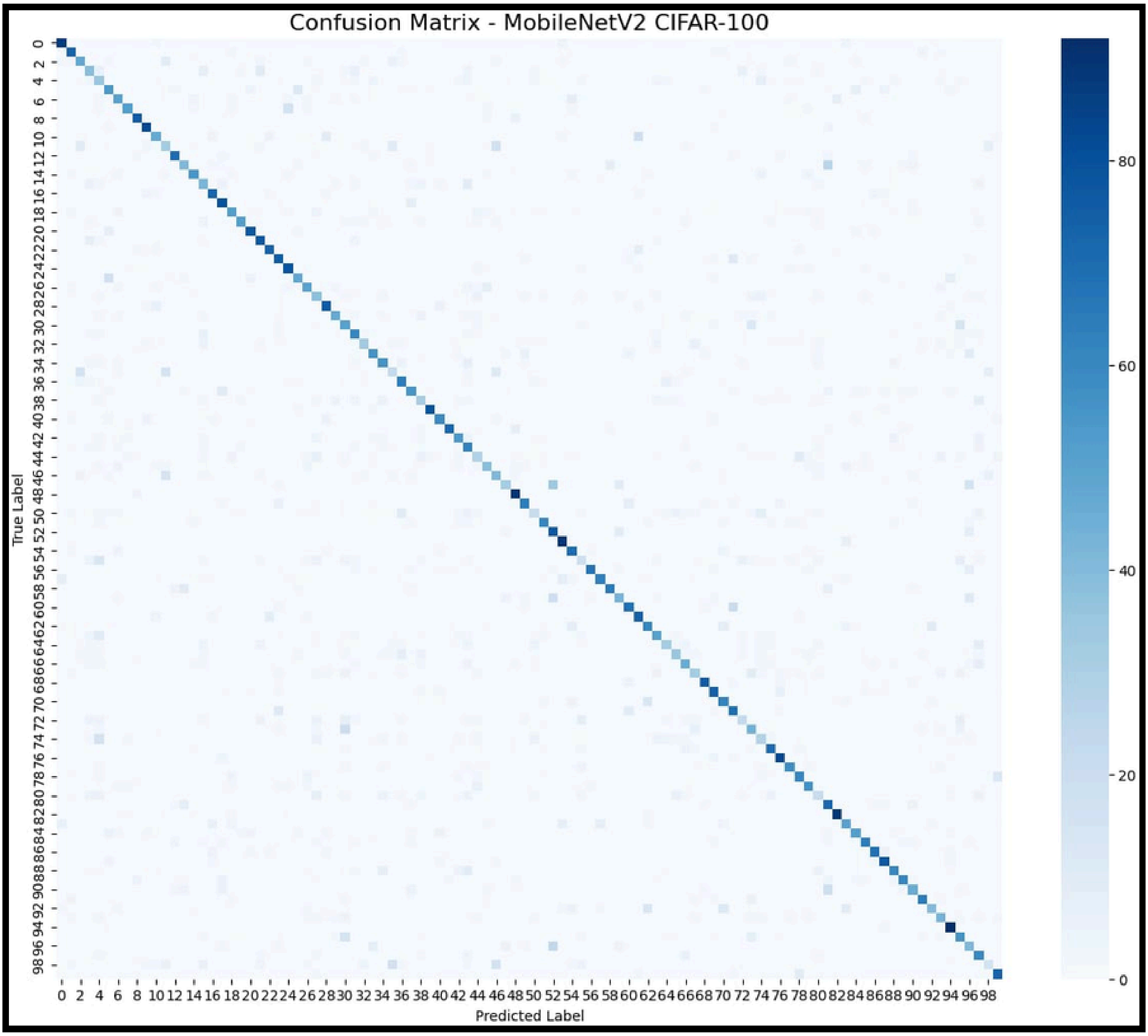
CONFUSION MATRIX

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==========================================================================
CONFUSION MATRIX SUMMARY
==========================================================================
Matrix Shape       : (100, 100)
Total Classes      : 100
Total Test Images : 10000
Correct Predictions : 5648
Wrong Predictions   : 4352
Test Accuracy       : 56.48%
==========================================================================
```

EXTERNAL IMAGE TESTING SUMMARY

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<Figure size 600x500 with 0 Axes>
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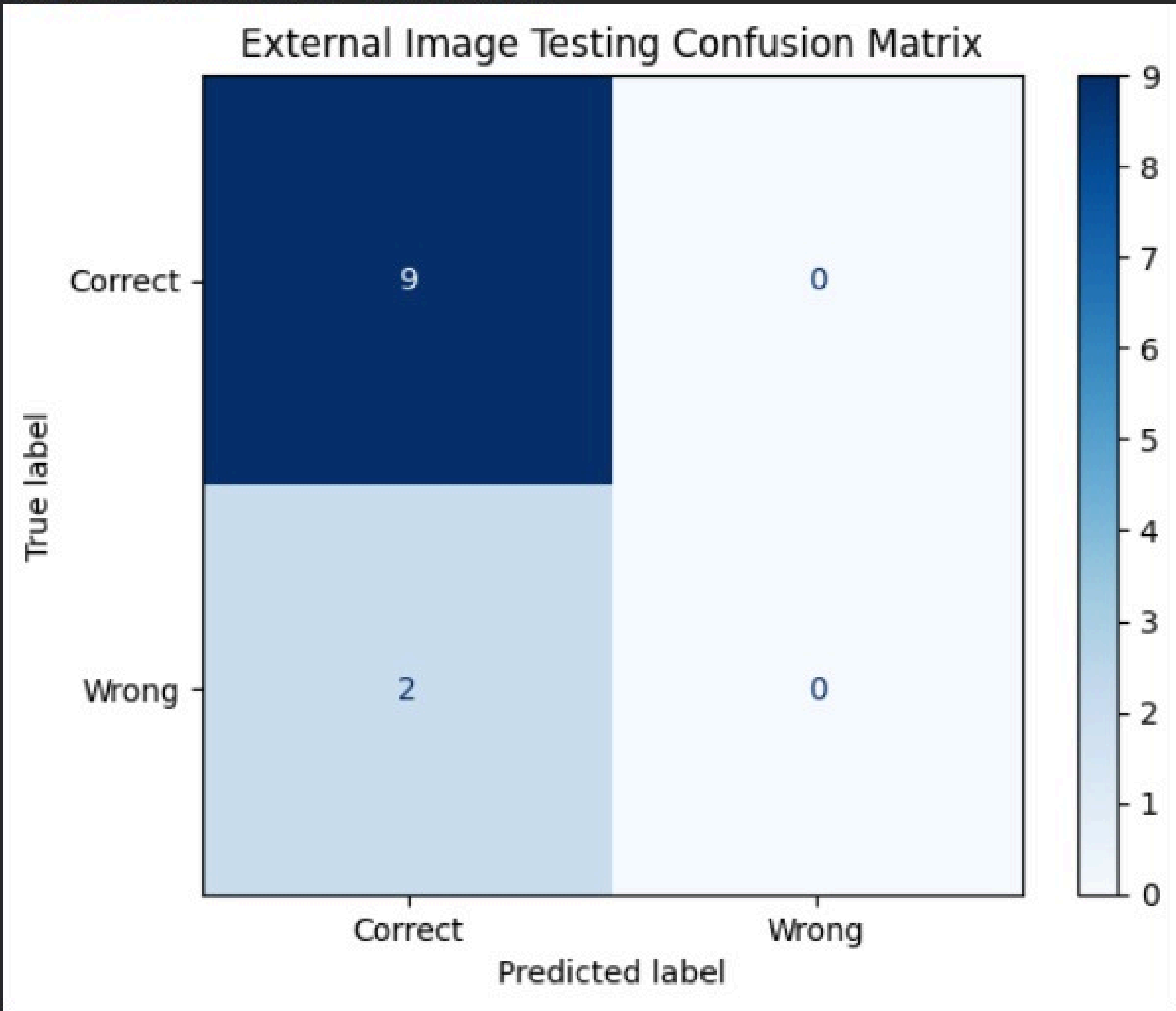


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=====================================================================
EXTERNAL IMAGE TESTING CONFUSION MATRIX SUMMARY
=====================================================================
Total Images Tested : 11
Correct Predictions : 9
Wrong Predictions   : 2
External Accuracy   : 81.82%
=====================================================================
```

CONFUSION MATRIX SUMMARY

```
============================================================
CONFUSION MATRIX SUMMARY
============================================================

• Matrix Shape          : (100, 100)
• Total Test Images     : 10000
• Correct Predictions   : Approximately 5648
• Wrong Predictions     : Approximately 4352
• Overall Accuracy      : 56.48%

The confusion matrix shows that the model correctly
classified a substantial number of CIFAR-100 test images.
Most errors occurred between visually similar classes,
which is expected in a fine-grained 100-class dataset.

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TRAINING CURVE SUMMARY
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Accuracy Curve Analysis:

• Training Accuracy increased steadily throughout training.
• Validation Accuracy improved consistently and reached
  approximately 56.34%.
• Early Stopping prevented unnecessary overfitting.
• The model demonstrated stable convergence.

Loss Curve Analysis:

• Training Loss decreased continuously during training.
• Validation Loss reduced significantly and stabilized.
• No severe divergence between training and validation
  loss was observed.
• The model achieved good generalization performance.

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EXTERNAL IMAGE TESTING ANALYSIS
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• Images Tested         : 11
• Correct Predictions   : 9
• Wrong Predictions     : 2
• External Accuracy     : 81.82%

The model successfully generalized to unseen images
downloaded from external sources and achieved high
prediction confidence on correctly classified samples.
```